AgriSoko Pocket G11 - Project Report

Group G11 | Version 1.0.0 | Repository: cxnisme/agrisoko-pocket-g11 | 23 October 2026

Project overview

AgriSoko Pocket G11 is an Expo/React Native Android application paired with a FastAPI member lookup service. It supports task-based member lookup, explainable profile matching, and persistent offline fallback. The directory presents GitHub usernames and assigned task areas.

Team leadership and authorship

This is our group project, led by Christian (cxnisme) as group leader. Christian integrated the mobile app, matching logic, API, and offline behavior with assistance from an AI coding assistant for implementation, focused tests, review, and documentation.

System design

The mobile client uses typed bottom-tab navigation, query validation, request timeouts, source badges, and AsyncStorage caching for the directory, profiles, and successful queries. When the API cannot be reached, bundled task records and local token matching provide offline lookup. The API exposes health, model-info, member-directory, profile, and ask endpoints. Matching uses deterministic TF-IDF/cosine-style term weighting with exact name and alias boosts plus confidence and margin checks.

Recorded automated validation

Area	Recorded check	Result
Backend	python -m pytest -q (Python 3.12)	13 passed
Mobile	npm test	7 passed
Mobile	npm run typecheck	Passed
Expo	npx expo install --check	Dependencies up to date
Expo	npx expo-doctor	18/18 checks passed
Android bundle	npx expo export --platform android	Succeeded

Production APK

EAS build 51746a29-700a-438c-b3eb-3357ddd7fd9a succeeded on 23 October 2026 using app version 1.0.0 (version code 1), Expo SDK 53.0.0, and source revision 606779e. The APK is included as build/AgriSokoPocket_G11.apk. Its SHA-256 is AD64418C85B1B7AA56DCF3FC5F5DBF9AFA579038E4DCCC64C2343DEFA398F2A8. Archive inspection found 1,065 entries, including AndroidManifest.xml, classes.dex, resources.arsc, and native libraries.

Release status and remaining evidence

The app bundle contains the Android emulator API fallback http://10.0.2.2:8000; no hosted API URL was embedded in this APK. Live API access from physical devices therefore needs a deployed HTTPS API URL, a replacement EAS APK, and device validation. No physical-device test, demo recording, external user feedback, or member profile sign-off is included. The matcher evaluation artifact is also not present.

Submission package

The submission package contains the source, project report, contribution record, automated test log, feedback status, AI-use disclosure, and the verified APK. The demo recording is not available and is not represented as completed.